Applied Statistical Modeling 2

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Regression models for constrained, censored and truncated

Constrained data are represented by variables, whose values are constrained into a specific interval

$$y \in [a, b] \quad a, b \in R$$

Regression models for constrained, censored and truncated

An example of constrained data, which is frequent in many practical applications is represented by variables, whose values are constrained in the interval [0,1]

$$y \in [0, 1]$$

as it is the case, for exmple, for variables representing proportions (i.e., proportion of consumption of certain goods, proportion of expenditure on specific goods)

Regression models for constrained, censored and truncated

Let's have a general definition of censored, truncated and constrained data, generalizing the definitions we've seen in the context of time-to-event data

Regression models for constrained, censored and truncated

Censoring generally occurs when we can observe the true value of a dependent variable of interest Y, only in a specific range of observations, and values below or above a certain range are "clustered" on a single value

- $y = y^*$ for all $y < y^*$ *right-censored*, or
- $y = y^*$ for all $y > y^*$ *left-censored*

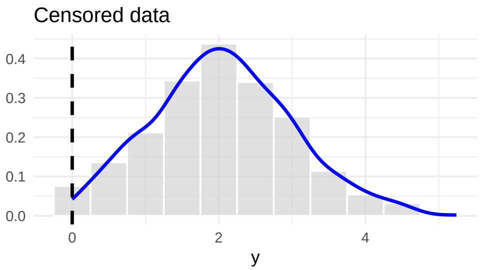
Regression models for constrained, censored and truncated

An example is the evaluation of household income, when the collection of income information, for exogenous reasons, is something like:

- household income>50,000 Euro, or
- household income $\leq$ 15,000 Euro
- ...

In any case, when censoring occurs, even if we are not able to observe the complete distribution of values for Y, we can observe y for all the n observations, so we have all the couples (Y, X)

Regression models for constrained, censored and truncated



Regression models for constrained, censored and truncated

Truncation occurs when, for a given censored dependent variable of interest Y, and a set of independent variables X, we do not have information on all the couples (Y, X), i.e., we do not have information about the type of job of the head of the household for those families whose income is below 15,000 Euro In this case, we are not able to have complete information about the probability density function of Y

Beta regression

When dealing with variables constrained in the interval $[0, 1]$ the standard linear regression could not be applied as

- it may yield estimated values outside the interval
- even if the variables are transformed to cover a different interval, often variables are asymmetrical and the simple linear regression model doesn't work well

Beta regression

To properly take into account the constrained nature of the variable is possible to use models based on the **beta** distribution

The Beta density is given by

$$B(\alpha,\beta) \sim \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}$$

where $\Gamma()$ is the gamma function and both α and β are shape parameters with, $\alpha, \beta > 0$

Beta regression

For $x \sim B(\alpha, \beta)$

$$E[X] = \frac{\alpha}{\alpha + \beta}$$

and

$$var[X] = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

Beta regression

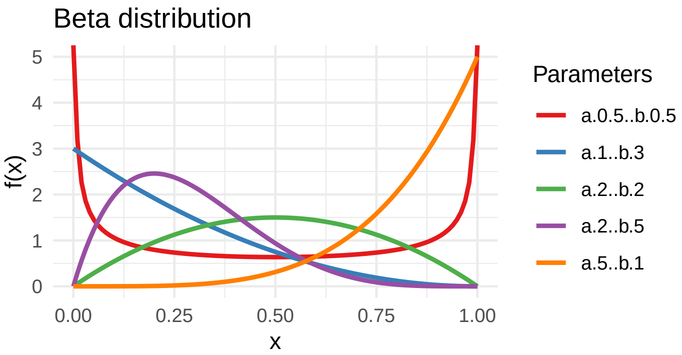
The Beta distribution is very flexible according to possible values of α and β

$$E[X] = \frac{\alpha}{\alpha + \beta}$$

and

$$var[X] = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

Regression models for constrained, censored and truncated



Beta regression

To formulate a model for a dependent variable Y that follows a Beta distribution, we need to define a regression structure for the mean of the dependent variable and a precision parameter using a different parametrization of the beta density, let

$$\mu = \frac{\alpha}{\alpha + \beta}$$

and

$$\phi = \alpha + \beta$$

Beta regression

According to this parametrization

$$E(y) = \mu$$

and

$$var(y) = \frac{V(\mu)}{1+\phi}$$

where $V(\mu) = \mu(1-\mu)$, so that μ is the mean of the response variable and ϕ can be interpreted as a precision parameter as for fixed μ, the larger the value of ϕ, the smaller the variance of y

Beta regression

The density function of y could thus be written as

$$f(y; \mu, \phi) = \frac{\Gamma(\phi)}{\Gamma(\mu\phi)\Gamma((1-\mu)\phi)} y^{\mu\phi - 1}(1 - y)^{(1-\mu)\phi - 1}$$

where $0 < \mu < 1$ and $\phi > 0$

Beta regression

The regression model for Y according to explanatory variable X could then be specified as follows

$$g(\mu) = \sum x_i \beta_i$$

where $g()$ is a strictly monotonic link function that maps (0,1) into R

Beta regression

There are different possible choices of the $g()$ function, one particularly useful is the logit

$$g(\mu) = log(\frac{\mu}{1-\mu})$$

so that $exp(\beta)$ could be easily interpreted in terms of odds

Beta regression

Parameters estimate could be obtained using the maximum likelihood (ML) method using a non-linear optimization algorithm (i.e., Newton algorithm)

The regression parameters are interpretable in terms of the mean of (the variable of interest), and the model is naturally heteroskedastic and accomodates asymmetries

Beta regression

Goodness of fit of the model could be assessed using the Pseudo R^2, defined as the square of the correlation between $x\hat{\beta}$ and $g(y)$, considering that $0 \leq R_p^2 \leq 1$, and whit $R_p^2 = 1$ implying perfect correlation between $x\hat{\beta}$ and $g(y)$ and thus between μ and $g(y)$